

60016 OPERATIONS RESEARCH

Linear Programs in Standard Form

Lecture 02

Last Lecture

LP as a tool for optimal decision making:

- ▶ Maximise/minimise a linear objective function
 - ▶ Linear constraints (equalities and/or inequalities)
 - ▶ The feasible region is a convex polyhedron
 - ▶ The vertices of the feasible region contain a solution to the LP problem (if the LP is well-defined)
- ⇒ An LP can be solved by examining all vertices, but this approach is computationally prohibitive!

This Lecture

- ▶ How to formulate an LP in a **standard way**

LPs in Standard Form

We want to use **computers** to solve LP problems

⇒ We need a **standardised specification** of LP problems



Definition: An LP is in **standard form** if:

- ▶ The aim is to **minimise a linear objective function**;
- ▶ All constraints are **linear equality constraints**;
- ▶ All **constraint right hand sides are non-negative**;
- ▶ All **decision variables are non-negative**.

LPs in Standard Form

An LP in standard form looks as follows:

$$\begin{array}{llllllllll} \text{minimise} & z = & c_1x_1 & + & c_2x_2 & + & \dots & + & c_nx_n & \\ \text{subject to} & & a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & = & b_1 \\ & & a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & = & b_2 \\ & & \vdots & & \vdots & & & & \vdots & & \vdots \\ & & a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & = & b_m \\ & & & & & & & & & & \\ & & & & & & & & & & x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0 \end{array}$$

The input parameters b_i , c_j , and a_{ij} are fixed real constants that encode the LP problem. We require $b_i \geq 0$, $\forall i = 1, \dots, m$. (The decision variables x_i , $i = 1, \dots, n$, are yet to be found.)

Compact Notation

Collect the **input parameters** in **vectors and matrices**:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad c = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$$c^T = [c_1, c_2, \dots, c_n]$$

LPs in Standard Form (cont)

- ▶ With matrix notation, the LP in standard form reduces to

$$\begin{array}{ll}\text{minimise} & z = c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0,\end{array}$$

where $b \geq 0$.

- ▶ Inequalities of the type $x \geq 0$ are understood to hold **component-wise**, i.e., $x_i \geq 0$, $\forall x_i \in x$.

Mentimeter 1: is this LP in standard form?

$$\begin{array}{ll}\text{minimise} & z = x_1 + 5x_2 - 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 = 3 \\ & x_1 + x_3 = 3 \\ & x_1, x_2 \geq 0\end{array}$$

Mentimeter 2: is this LP in standard form?

$$\begin{array}{ll}\text{minimise} & z = x_1 + 5x_2 - 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 = 3 \\ & x_1 + x_3 = 3 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

Mentimeter 3: is this LP in standard form?

$$\begin{array}{ll}\text{minimise} & z = x_1 + 5x_2 - 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 = -3 \\ & x_1 + x_3 = 3 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

Mentimeter 4: is this LP in standard form?

$$\begin{array}{ll}\text{minimise} & z = x_1 + 5x_2 - 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 \geq 3 \\ & x_1 + x_3 = 3 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

Standardising General LPs

General LP problems can

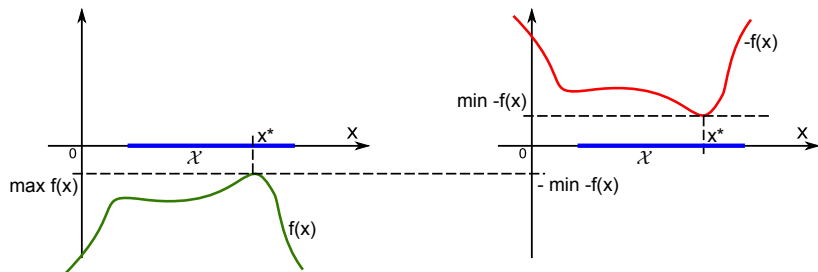
- ▶ be **maximisation** (instead of minimisation) problems;
- ▶ have **inequality** (instead of equality) **constraints**;
- ▶ have equality constraints with **negative** (instead of non-negative) **right hand sides**;
- ▶ have **free** (instead of non-negative) **decision variables**.

These general LPs can be transformed to standard LPs **in a systematic way**.

Maximisation $\rightarrow$ Minimisation

$$\left. \begin{array}{ll} \max & y = f(x) \\ \text{s.t.} & x \in \mathcal{X} \end{array} \right\} = \left\{ \begin{array}{ll} -\min & z = -f(x) \\ \text{s.t.} & x \in \mathcal{X} \end{array} \right.$$

Inverting the objective **preserves the optimal solution** x^*



Optimal value of the objective is $y^* = -z^*$.

$\leq$ Inequalities $\rightarrow$ Equalities

$$\text{minimise } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

subject to:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & \leq & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & \leq & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & \leq & b_m \end{array}$$

and

$$x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$$

$\leq$ Inequalities $\rightarrow$ Equalities

$$\text{minimise } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

subject to:

$$\begin{array}{ccccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & + & s_1 & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & + & s_2 & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & + & s_m & = & b_m \end{array}$$

and

$$x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0, \quad s_1 \geq 0, s_2 \geq 0, \dots, s_m \geq 0$$

Slack Variables

- ▶ To reformulate $\leq$ inequalities as equalities, we introduced m slack variables
 - ▶ Original variables: $x_1, x_2, \dots, x_n$
 - ▶ Slack variables: $s_1, s_2, \dots, s_m$ $\Rightarrow$ After transformation, LP has $n + m$ variables!
- ▶ With matrix notation we can write

$$\left. \begin{array}{ll} \min & z = c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array} \right\} = \left\{ \begin{array}{ll} \min & z = c^T x \\ \text{s.t.} & Ax + s = b \\ & x \geq 0, s \geq 0, \end{array} \right.$$

where $s = (s_1, \dots, s_m)^T$.

- ▶ Slack variables take the value of the difference $b - Ax$

$\geq$ Inequalities $\rightarrow$ Equalities

$$\text{minimise } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

subject to:

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & \geq & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & \geq & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & \geq & b_m \end{array}$$

and

$$x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$$

$\geq$ Inequalities $\rightarrow$ Equalities

$$\text{minimise } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

subject to:

$$\begin{array}{ccccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & - & s_1 & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & - & s_2 & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & - & s_m & = & b_m \end{array}$$

and

$$x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0, \quad s_1 \geq 0, s_2 \geq 0, \dots, s_m \geq 0$$

Excess Variables

- ▶ To reformulate $\geq$ inequalities as equalities, we introduced m **excess variables** (a.k.a. surplus variables)
 - ▶ **Original variables**: $x_1, x_2, \dots, x_n$
 - ▶ **Excess variables**: $s_1, s_2, \dots, s_m$ $\Rightarrow$ After transformation, LP has $n + m$ variables!
- ▶ With matrix notation we can write

$$\left. \begin{array}{ll} \min & z = c^T x \\ \text{s.t.} & Ax \geq b \\ & x \geq 0 \end{array} \right\} = \left\{ \begin{array}{ll} \min & z = c^T x \\ \text{s.t.} & Ax - s = b \\ & x \geq 0, s \geq 0, \end{array} \right.$$

where $s = (s_1, \dots, s_m)^T$.

- ▶ Excess variables take the value of the difference $Ax - b$

Equivalence

- ▶ Assume initial problem **not** in standard form
- ▶ x : feasible solution to the initial problem
- ▶ (x, s) : feasible solution to the standardised problem
- ▶ x can be associated with **one and only one** (x, s) using the reformulations we have defined.
- ▶ In particular, the optimal solutions will be x^* and (x^*, s^*)
 - ▶ x^* will be the same in both formulations

Negative Right Hand Sides

- ▶ If the right hand side of the i th constraint is **negative**, i.e., if $b_i < 0$ in

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i,$$

then this constraint should be **multiplied by -1** .

- ▶ This yields

$$(-a_{i1})x_1 + (-a_{i2})x_2 + \dots + (-a_{in})x_n = -b_i.$$

- ▶ The new constraint has a **non-negative right hand side**, i.e., we have $-b_i \geq 0$.

Free Variables (1st Approach)

Free variables:

- ▶ Suppose there is no constraint $x_j \geq 0$, i.e., x_j can be positive or negative.
- ▶ Substitute $x_j = x_j^+ - x_j^-$ with $x_j^+, x_j^- \geq 0$.
- ▶ The LP has now $(n + 1)$ variables:

$$x_1, \dots, x_{j-1}, x_j^+, x_j^-, x_{j+1}, \dots, x_n$$

Free Variables (2nd Approach)

Free variables:

- ▶ Suppose there is no constraint $x_j \geq 0$, i.e., x_j can be positive or negative.
- ▶ Any equality constraint involving x_j can be used to eliminate x_j .
- ▶ Example: x_1 is free

$$\left. \begin{array}{ll} \min & z = x_1 + 3x_2 + 4x_3 \\ \text{s.t.} & x_1 + 2x_2 + x_3 = 5 \quad (*) \\ & 2x_1 + 3x_2 + x_3 = 6 \\ & x_2, x_3 \geq 0 \end{array} \right\} = \left\{ \begin{array}{ll} \min & z = x_2 + 3x_3 + 5 \\ \text{s.t.} & x_2 + x_3 = 4 \\ & x_2, x_3 \geq 0 \end{array} \right.$$

Use (*) to substitute $x_1 = 5 - 2x_2 - x_3$.